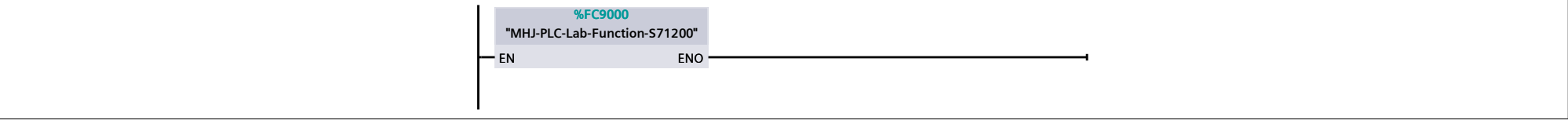


Main [OB1]

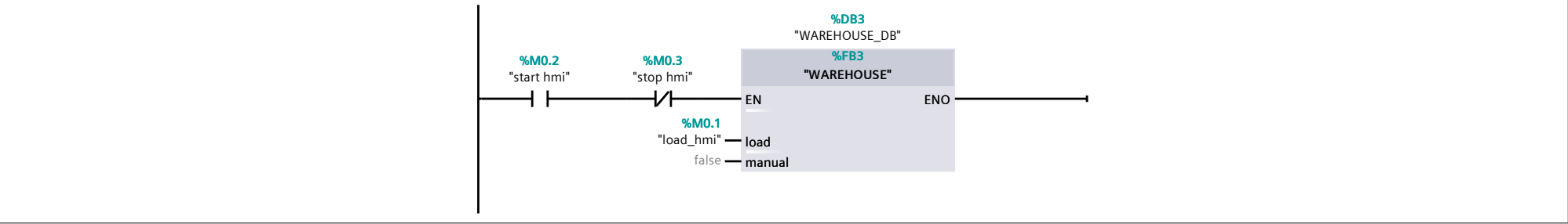
| Main Properties |                              |                 |   |         |    |          |     |
|-----------------|------------------------------|-----------------|---|---------|----|----------|-----|
| General         |                              |                 |   |         |    |          |     |
| Name            | Main                         | Number          | 1 | Type    | OB | Language | LAD |
| Numbering       | Automatic                    |                 |   |         |    |          |     |
| Information     |                              |                 |   |         |    |          |     |
| Title           | "Main Program Sweep (Cycle)" | Author          |   | Comment |    | Family   |     |
| Version         | 0.1                          | User-defined ID |   |         |    |          |     |

| Name         | Data type | Default value | Comment                               |
|--------------|-----------|---------------|---------------------------------------|
| ▼ Input      |           |               |                                       |
| Initial_Call | Bool      |               | Initial call of this OB               |
| Remanence    | Bool      |               | =True, if remanent data are available |
| Temp         |           |               |                                       |
| Constant     |           |               |                                       |

Network 1:



Network 2:



load [FB2]

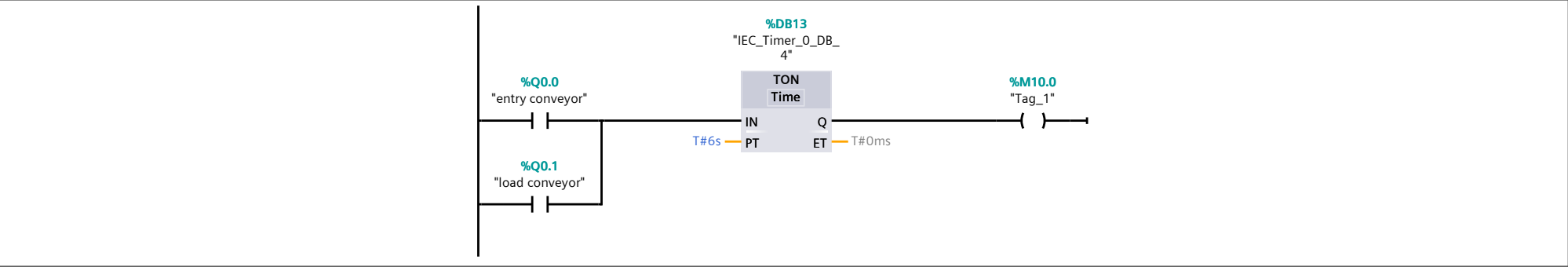
| load Properties |           |                 |   |         |    |          |     |
|-----------------|-----------|-----------------|---|---------|----|----------|-----|
| General         |           |                 |   |         |    |          |     |
| Name            | load      | Number          | 2 | Type    | FB | Language | LAD |
| Numbering       | Automatic |                 |   |         |    |          |     |
| Information     |           |                 |   |         |    |          |     |
| Title           |           | Author          |   | Comment |    | Family   |     |
| Version         | 0.1       | User-defined ID |   |         |    |          |     |

| Name     | Data type | Default value | Retain | Accessible from HMI/OPC UA/Web API | Writ-able from HMI/OPC UA/ Web API | Visible in HMI engi-neering | Setpoint | Supervi-sion | Comment |
|----------|-----------|---------------|--------|------------------------------------|------------------------------------|-----------------------------|----------|--------------|---------|
| Input    |           |               |        |                                    |                                    |                             |          |              |         |
| Output   |           |               |        |                                    |                                    |                             |          |              |         |
| InOut    |           |               |        |                                    |                                    |                             |          |              |         |
| Static   |           |               |        |                                    |                                    |                             |          |              |         |
| Temp     |           |               |        |                                    |                                    |                             |          |              |         |
| Constant |           |               |        |                                    |                                    |                             |          |              |         |

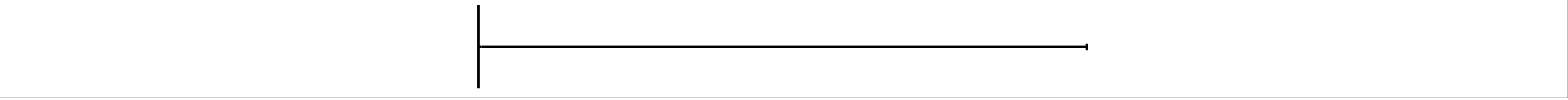
Network 1:



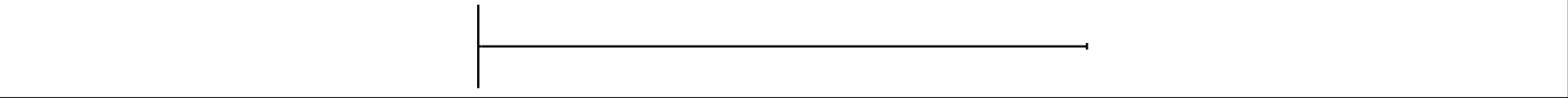
Network 2:



Network 3:



Network 4:



sort [FB4]

| sort Properties |           |                 |   |         |    |          |     |
|-----------------|-----------|-----------------|---|---------|----|----------|-----|
| General         |           |                 |   |         |    |          |     |
| Name            | sort      | Number          | 4 | Type    | FB | Language | LAD |
| Numbering       | Automatic |                 |   |         |    |          |     |
| Information     |           |                 |   |         |    |          |     |
| Title           |           | Author          |   | Comment |    | Family   |     |
| Version         | 0.1       | User-defined ID |   |         |    |          |     |

| Name     | Data type | Default value | Retain | Accessible from HMI/OPC UA/Web API | Writ-able from HMI/OPC UA/ Web API | Visible in HMI engi-neering | Setpoint | Supervi-sion | Comment |
|----------|-----------|---------------|--------|------------------------------------|------------------------------------|-----------------------------|----------|--------------|---------|
| Input    |           |               |        |                                    |                                    |                             |          |              |         |
| Output   |           |               |        |                                    |                                    |                             |          |              |         |
| InOut    |           |               |        |                                    |                                    |                             |          |              |         |
| Static   |           |               |        |                                    |                                    |                             |          |              |         |
| Temp     |           |               |        |                                    |                                    |                             |          |              |         |
| Constant |           |               |        |                                    |                                    |                             |          |              |         |

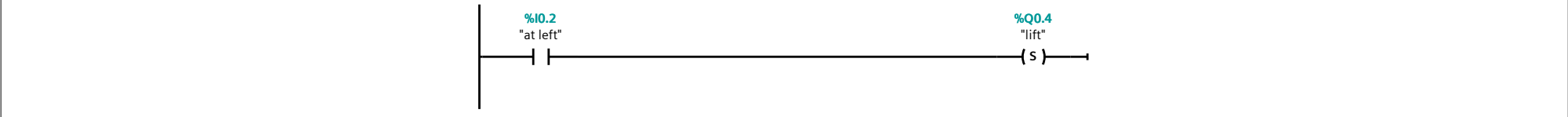
Network 1:



Network 2:



Network 3:



state machine [FB1]

| state machine Properties |               |                 |   |         |    |          |     |
|--------------------------|---------------|-----------------|---|---------|----|----------|-----|
| General                  |               |                 |   |         |    |          |     |
| Name                     | state machine | Number          | 1 | Type    | FB | Language | SCL |
| Numbering                | Automatic     |                 |   |         |    |          |     |
| Information              |               |                 |   |         |    |          |     |
| Title                    |               | Author          |   | Comment |    | Family   |     |
| Version                  | 0.1           | User-defined ID |   |         |    |          |     |

| Name     | Data type | Default value | Retain     | Accessible from HMI/OPC UA/Web API | Writ-able from HMI/OPC UA/ Web API | Visible in HMI engi-neering | Setpoint | Supervi-sion | Comment |
|----------|-----------|---------------|------------|------------------------------------|------------------------------------|-----------------------------|----------|--------------|---------|
| Input    |           |               |            |                                    |                                    |                             |          |              |         |
| Output   |           |               |            |                                    |                                    |                             |          |              |         |
| InOut    |           |               |            |                                    |                                    |                             |          |              |         |
| ▼ Static |           |               |            |                                    |                                    |                             |          |              |         |
| process1 | Bool      | false         | Non-retain | True                               | True                               | True                        | False    |              |         |
| process2 | Bool      | false         | Non-retain | True                               | True                               | True                        | False    |              |         |
| step     | Byte      | 16#0          | Non-retain | True                               | True                               | True                        | False    |              |         |
| process3 | Bool      | false         | Non-retain | True                               | True                               | True                        | False    |              |         |
| process4 | Bool      | false         | Non-retain | True                               | True                               | True                        | False    |              |         |
| Temp     |           |               |            |                                    |                                    |                             |          |              |         |
| Constant |           |               |            |                                    |                                    |                             |          |              |         |

```
0001
0002 CASE #step OF
0003   0:
0004     IF "storage_DB".out = 54 THEN
0005       #process1 := FALSE;
0006       #process2 := FALSE;
0007       #process3 := FALSE;
0008       #step := 0;
0009
0010     ELSIF NOT "at entry" THEN
0011       #process1 := TRUE;
0012       #process2 := FALSE;
0013       #process3 := FALSE;
0014       #step := 1;
0015     END_IF;
0016
0017   ;
0018   1:
0019     IF NOT "at load" THEN
0020       #process1 := FALSE;
0021       #process2 := TRUE;
0022       "IEC_Timer_0_DB_4".IN := FALSE;
0023       #step := 2;
0024     ;
0025     END_IF;
0026   2:
0027     IF "lift" AND NOT "at left" THEN
0028       #process2 := FALSE;
0029       #process3 := TRUE;
0030       #step := 3;
0031
0032     ;
0033     END_IF
0034
0035   ;
0036   3:
0037     IF  "IEC_Timer_0_DB_2".Q THEN
0038       #process3 := FALSE;
0039       "target position" := 101;
0040       #step := 0;
0041       "storage_DB".found := false;
0042
0043     ;
0044     END_IF;
0045
0046   ;
0047 END_CASE;
0048
```

| Symbol                | Address | Type | Comment |
|-----------------------|---------|------|---------|
| "at entry"            | %I0.0   | Bool |         |
| "at left"             | %I0.2   | Bool |         |
| "at load"             | %I0.1   | Bool |         |
| "IEC_Timer_0_DB_2".Q  |         | Bool |         |
| "IEC_Timer_0_DB_4".IN |         | Bool |         |
| "lift"                | %Q0.4   | Bool |         |

|                                      |         |       |         |  |
|--------------------------------------|---------|-------|---------|--|
| Totally Integrated Automation Portal |         |       |         |  |
| Symbol                               | Address | Type  | Comment |  |
| "storage_DB".found                   |         | Bool  |         |  |
| "storage_DB".out                     |         | DWord |         |  |
| "target position"                    | %QD30   | DWord |         |  |
| #process1                            |         | Bool  |         |  |
| #process2                            |         | Bool  |         |  |
| #process3                            |         | Bool  |         |  |
| #step                                |         | Byte  |         |  |
|                                      |         |       |         |  |
|                                      |         |       |         |  |

Totally Integrated Automation Portal

storage [FB5]

storage Properties

General

Name

storage

Number

5

Type

FB

Language

SCL

Numbering

Automatic

Information

Title

Author

Comment

Family

Version

0.1

User-defined ID

| Name        | Data type            | Default value | Retain     | Accessible from HMI/OPC UA/Web API | Writ-able from HMI/ OPC UA/ Web API | Visible in HMI engi-neering | Setpoint | Supervi-sion | Comment |
|-------------|----------------------|---------------|------------|------------------------------------|-------------------------------------|-----------------------------|----------|--------------|---------|
| Input       |                      |               |            |                                    |                                     |                             |          |              |         |
| ▼ Output    |                      |               |            |                                    |                                     |                             |          |              |         |
| out         | DWord                | 16#0          | Non-retain | True                               | True                                | True                        | False    |              |         |
| InOut       |                      |               |            |                                    |                                     |                             |          |              |         |
| ▼ Static    |                      |               |            |                                    |                                     |                             |          |              |         |
| ▼ storage   | Array[1..54] of Bool |               | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[1]  | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[2]  | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[3]  | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[4]  | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[5]  | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[6]  | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[7]  | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[8]  | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[9]  | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[10] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[11] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[12] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[13] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[14] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[15] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[16] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[17] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[18] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[19] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[20] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[21] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[22] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[23] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[24] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[25] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[26] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[27] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[28] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[29] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[30] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[31] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[32] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[33] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[34] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[35] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[36] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[37] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[38] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[39] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[40] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[41] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[42] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[43] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[44] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[45] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[46] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[47] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[48] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[49] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[50] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[51] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[52] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[53] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| storage[54] | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |
| found       | Bool                 | false         | Non-retain | True                               | True                                | True                        | False    |              |         |

Totally Integrated Automation Portal

| Name     | Data type | Default value | Retain | Accessible from HMI/OPC UA/Web API | Writ-able from HMI/OPC UA/ Web API | Visible in HMI engi-neering | Setpoint | Supervi-sion | Comment |
|----------|-----------|---------------|--------|------------------------------------|------------------------------------|-----------------------------|----------|--------------|---------|
| ▼ Temp   |           |               |        |                                    |                                    |                             |          |              |         |
| i        | Int       |               |        |                                    |                                    |                             |          |              |         |
| j        | Int       |               |        |                                    |                                    |                             |          |              |         |
| Constant |           |               |        |                                    |                                    |                             |          |              |         |

0001

0002 FOR #i := 1 TO 54 DO

0003

0004

0005 IF NOT #storage[#i] AND NOT #found THEN

0006 #out := INT\_TO\_DWORD (#i);

0007 #storage[#i] := TRUE;

0008 #found := TRUE;

0009 ;

0010 END\_IF;

0011

0012

0013 END\_FOR;

0014

| Symbol      | Address | Type  | Comment |
|-------------|---------|-------|---------|
| #found      |         | Bool  |         |
| #i          |         | Int   |         |
| #out        |         | DWord |         |
| #storage[*] |         | Bool  |         |

store [FB6]

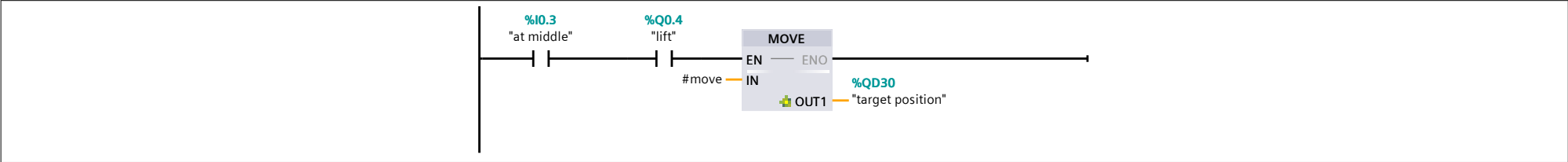
| store Properties |           |                 |   |         |    |          |     |
|------------------|-----------|-----------------|---|---------|----|----------|-----|
| General          |           |                 |   |         |    |          |     |
| Name             | store     | Number          | 6 | Type    | FB | Language | LAD |
| Numbering        | Automatic |                 |   |         |    |          |     |
| Information      |           |                 |   |         |    |          |     |
| Title            |           | Author          |   | Comment |    | Family   |     |
| Version          | 0.1       | User-defined ID |   |         |    |          |     |

| Name     | Data type | Default value | Retain     | Accessible from HMI/OPC UA/Web API | Writ-able from HMI/OPC UA/ Web API | Visible in HMI engi-neering | Setpoint | Supervi-sion | Comment |
|----------|-----------|---------------|------------|------------------------------------|------------------------------------|-----------------------------|----------|--------------|---------|
| Input    |           |               |            |                                    |                                    |                             |          |              |         |
| Output   |           |               |            |                                    |                                    |                             |          |              |         |
| InOut    |           |               |            |                                    |                                    |                             |          |              |         |
| ▼ Static |           |               |            |                                    |                                    |                             |          |              |         |
| move     | DWord     | 16#0          | Non-retain | True                               | True                               | True                        | False    |              |         |
| Temp     |           |               |            |                                    |                                    |                             |          |              |         |
| Constant |           |               |            |                                    |                                    |                             |          |              |         |

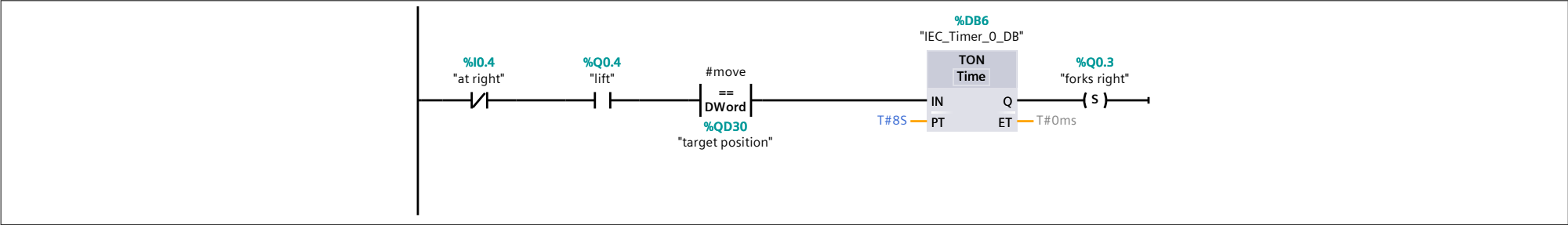
Network 1:



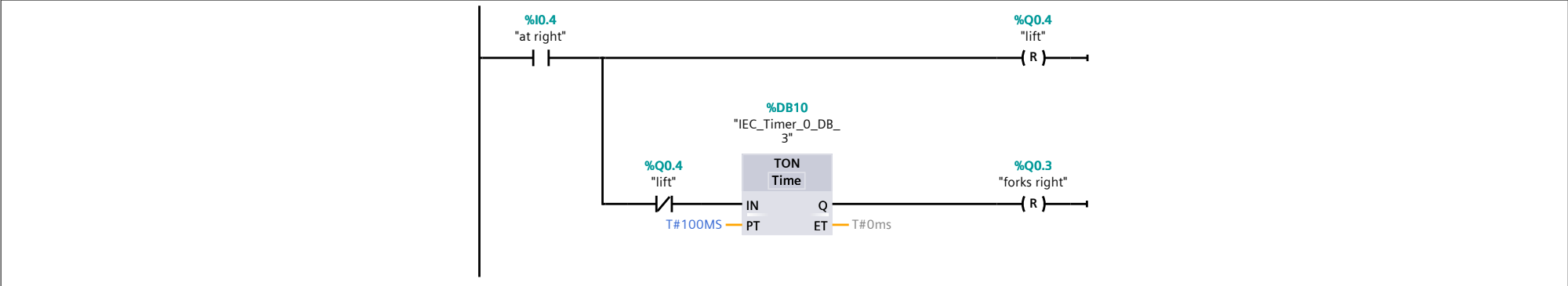
Network 2:



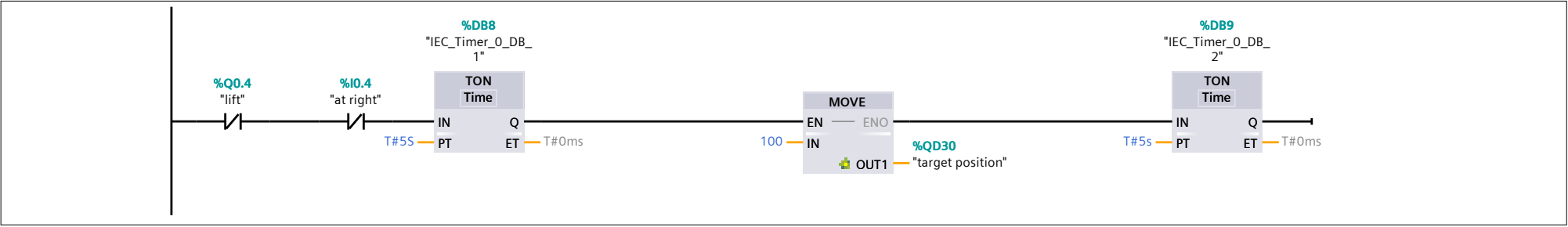
Network 3:



Network 4:



Network 5:

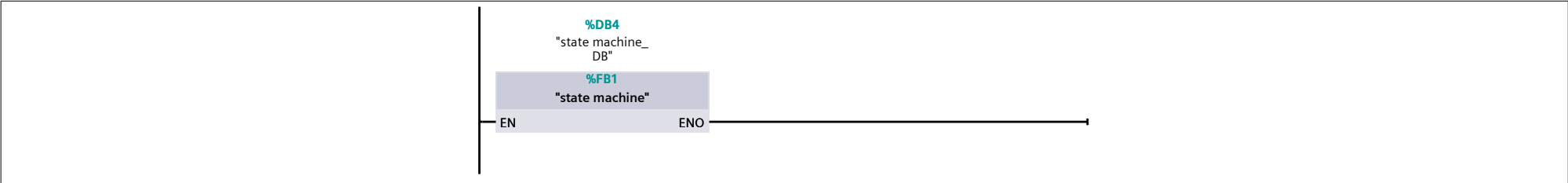


WAREHOUSE [FB3]

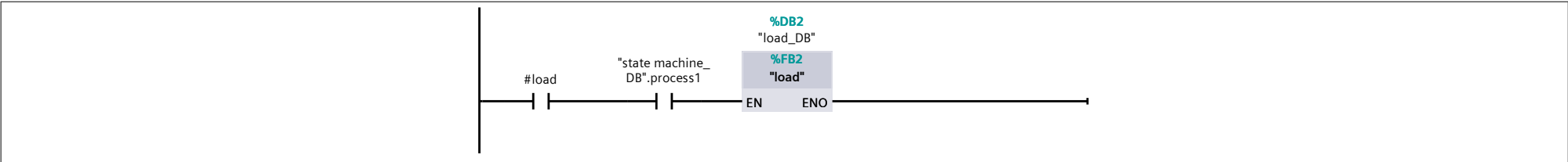
| WAREHOUSE Properties |           |                 |   |         |    |          |     |
|----------------------|-----------|-----------------|---|---------|----|----------|-----|
| General              |           |                 |   |         |    |          |     |
| Name                 | WAREHOUSE | Number          | 3 | Type    | FB | Language | LAD |
| Numbering            | Automatic |                 |   |         |    |          |     |
| Information          |           |                 |   |         |    |          |     |
| Title                |           | Author          |   | Comment |    | Family   |     |
| Version              | 0.1       | User-defined ID |   |         |    |          |     |

| Name     | Data type | Default value | Retain     | Accessible from HMI/OPC UA/Web API | Writ-able from HMI/OPC UA/ Web API | Visible in HMI engi-neering | Setpoint | Supervi-sion | Comment |
|----------|-----------|---------------|------------|------------------------------------|------------------------------------|-----------------------------|----------|--------------|---------|
| ▼ Input  |           |               |            |                                    |                                    |                             |          |              |         |
| load     | Bool      | false         | Non-retain | True                               | True                               | True                        | False    |              |         |
| manual   | Bool      | false         | Non-retain | True                               | True                               | True                        | False    |              |         |
| Output   |           |               |            |                                    |                                    |                             |          |              |         |
| InOut    |           |               |            |                                    |                                    |                             |          |              |         |
| Static   |           |               |            |                                    |                                    |                             |          |              |         |
| Temp     |           |               |            |                                    |                                    |                             |          |              |         |
| Constant |           |               |            |                                    |                                    |                             |          |              |         |

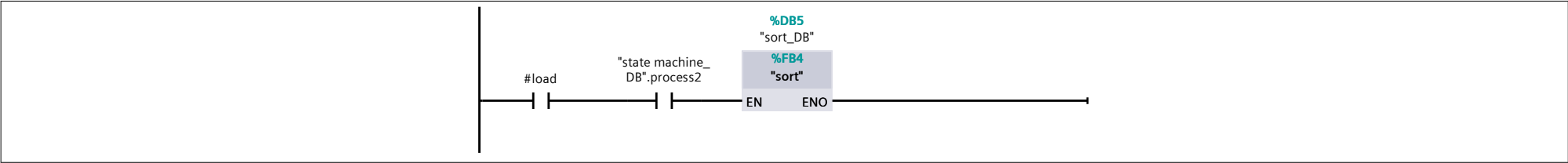
Network 1:



Network 2:



Network 3:



Network 4:

